Exercise 2

Q1] $S(x)$: x is a student
$P(x)$: x submitted a project

$$\forall x\,(S(x) \rightarrow P(x))$$

Q3] $S(x)$: x is a student
$P(x)$: x ~~is~~ submitted a project

$$\cancel{\neg \forall x\,(S(x) \rightarrow P(x))\ (\vee \neg \exists x\,(S(x) \wedge \neg P(x))}$$

Q3] $\exists x\,(S(x) \wedge \neg P(x))$

Q2] $S(x)$: x is a student
$P(x)$: x submitted a project

$$\exists x\,(S(x) \wedge P(x))$$

Q4] $S(x)$: x student
$C(x)$: x ~~is~~ cheated

$$\forall x\,(S(x) \rightarrow \neg C(x))\ (\vee\ \neg \exists x\,(S(x) \wedge C(x)))$$

Q5] $S(x)$: x is a student
$A(x)$: x ~~is~~ can access the lab

$$\forall x\,(A(x) \rightarrow S(x))$$

Q6] ~~$\forall(x)$~~ $L(x,y)$: x loves y

$$\forall x \exists y \, (L(x,y))$$

Q7] $L(x,y)$: x loves y

$$\exists y \forall x \, (L(x,y))$$

Q8] $A(x)$: x is admin
$R(x)$: x reset the server

$$\exists x \, (A(x) \land R(x) \land \forall y \, ((A(y) \land R(y)) \to x = y))$$

Q9] $S(x)$: x is student
$F(x)$: x failed the exam

$$\forall x \, \forall y \, ((S(x) \land F(x) \land S(y) \land F(y)) \to x = y)$$

Q10] $P(x)$ - x is a printer , $P(y)$ y is a printer
$O(x)$: x is online , $O(y)$ y is online

$$\cancel{\forall x \, (C(x) \to \exists y \, (C(y) \land P(y,x)))}$$

$$\exists x \exists y \, (P(x) \land P(y) \land O(x) \land O(y) \land x \neq y)$$

Q11) $C(x)$: x is a course

$P(x,y)$: x is a pre-req for y

$$\forall x\,(C(x) \rightarrow \exists y\,(C(y) \land P(y,x)))$$

Q12) $C(x)$: x is a course

$P(x,y)$: x is a pre-req for y

$$\exists x\,(C(x) \land \forall y\,(C(y) \rightarrow P(x,y)))$$

Q13) $P(x)$: x is a person

$A(x)$: x is a parent

$C(x,y)$: x is a child of y

$M(x,y)$: x admires y

$$\forall x\,((P(x) \land A(x)) \longrightarrow \exists y\,(P(y) \land C(y,x) \land M(y,x)))$$

Q14) $P(x)$: x is a person

$E(x,e)$: x has email e

$$\forall x \forall y \forall e\,((P(x) \land P(y) \land E(x,e) \land E(y,e)) \rightarrow x=y)$$

Q15) $E(x)$: x is even

$O(x)$: x is odd

$S(x)$: Successor function of x

$$\forall x\,(E(x) \longrightarrow O(S(x)))$$

Q16) $T(x)$: x is a team

$M(x,y)$: y is manager of x

$$\forall x\,(T(x) \longrightarrow \exists y\,(M(x,y) \land \forall z\,(M(x,z) \longrightarrow y=z)))$$

Q17) $P(x)$: x is a person

$M(x,y)$: x mentors y

$R(y,x)$: y respects x

$$\forall x\,\forall y\,((P(x) \land P(y) \land M(x,y)) \longrightarrow R(y,x))$$

Q18) $R(x)$: x is a researcher

$C(x,y)$: x cites y

$$\exists x\,(R(x) \wedge \forall y\,((R(y) \wedge \neg (x = y)) \rightarrow C(x,y)))$$

Q19) $P(p)$: p is a path from s to t

$V(v,p)$: v is a vertex in p

$S(v)$: v is labeled 'safe'

$$\forall p\,(P(p) \rightarrow \exists v\,(V(v,p) \wedge S(v)))$$

Q20) $N(x)$: x is a number

$G(x,y)$: x is a greater than y

$P(x)$: x is positive

$P(y,x)$: y is a predecessor of x

$$\forall x\,((N(x) \wedge \forall y\,(R(y,x) \rightarrow G(x,y))) \rightarrow P(x))$$